LEVERAGING ARTIFICIAL GENERAL INTELLIGENCE IN THE AVIATION INDUSTRY

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Introduction

Artificial intelligence is already changing the aviation industry through predictive maintenance, flight planning, customer service, security screening, weather analysis, and air traffic management. However, most AI systems used today are considered narrow AI because they are designed to complete one task or operate within one limited area. For example, one system may predict when an aircraft part will fail, while another analyzes weather or helps schedule airline crews. These systems can be highly effective, but they normally cannot transfer their knowledge from one area to another without additional programming or training.

Artificial General Intelligence, commonly called AGI, represents a much broader form of intelligence. AGI can be described as a system capable of learning, reasoning, planning, and applying knowledge across many different tasks and unfamiliar situations. Unlike narrow AI, an AGI system would not be limited to one function. It could examine a complex aviation problem involving aircraft maintenance, weather, crew availability, passenger needs, airport congestion, safety regulations, and environmental concerns at the same time. Researchers have proposed evaluating AGI based on its generality, performance, and level of autonomy, although there is still no universally accepted test for determining when AGI has been achieved (Morris et al., 2024). Because the term remains debated, this case study treats AGI as an emerging or future capability rather than a currently certified aviation technology.

AGI could have a significant impact on aviation because aviation is a highly connected industry. A decision made in one area can affect many other areas. A mechanical delay can change crew schedules, gate assignments, connecting flights, passenger accommodation, fuel use, and air traffic flow. An AGI system could understand these relationships and recommend a complete solution rather than treating each problem separately. However, because aviation is safety-critical, AGI would need strict human oversight, security controls, testing, certification, and clear limits on its authority.

Table 1

Comparison of Current AI and AGI in Aviation

Area	Current Narrow AI	Proposed AGI
Task range	Performs a specific task	Performs and connects many tasks
Learning	Learns within a defined dataset or problem	Transfers knowledge to new situations
Decision-making	Produces predictions or limited recommendations	Reasons across maintenance, safety, weather, staffing, and operations
System integration	Usually operates within one department	Coordinates information across the aviation system
Human role	Human reviews individual AI output	Human remains responsible for approving safety-critical decisions
Example	Predicting engine maintenance needs	Reorganizing an entire flight operation after a maintenance and weather disruption

Current State and Challenges of the Aviation Industry

Modern aviation is one of the safest and most technically advanced transportation systems in the world. Airlines, aircraft manufacturers, airports, regulators, air traffic controllers, maintenance organizations, and security agencies use large amounts of data to operate safely. The Federal Aviation Administration's Next Generation Air Transportation System modernized communication, navigation, surveillance, automation, and information management within the United States. The purpose of these improvements was to increase the safety, capacity, predictability, flexibility, efficiency, and resilience of aviation operations (Federal Aviation Administration [FAA], 2026).

Despite these advancements, aviation continues to face several challenges. Increasing air traffic creates congestion in airports and airspace. Airlines must respond to changing weather, equipment failures, runway closures, security concerns, crew availability, and passenger disruptions. Maintenance organizations must examine large amounts of sensor data and inspection information while following detailed regulatory requirements. Airports must manage baggage, gates, ground vehicles, security checkpoints, and emergency situations. The industry is also under pressure to reduce delays, control costs, strengthen cybersecurity, improve accessibility, and lower its environmental impact.

The International Civil Aviation Organization has identified AI as an important technology for processing aviation data, predicting congestion, supporting flight planning, identifying risks, and improving operational safety. Existing AI systems can analyze real-time weather, traffic, maintenance, and operational information. However, advanced automation may also affect situational awareness, decision-making, and the relationship between humans and machines. Aviation employees must therefore be trained to understand both the capabilities and limitations of automated systems (International Civil Aviation Organization [ICAO], 2025).

Sustainability is another major challenge. ICAO member states have established a long-term goal of achieving net-zero carbon emissions from international aviation by 2050. Reaching this goal will require improvements in aircraft technology, sustainable aviation fuels, airport infrastructure, air traffic procedures, and operational efficiency. Better coordination of flight routes, aircraft weight, weather conditions, ground operations, and fuel use could help reduce unnecessary emissions (ICAO, n.d.).

Existing Uses of AI in Aviation

The aviation industry already uses narrow AI and machine learning in several areas. Predictive-maintenance systems examine information from aircraft sensors and maintenance records to detect unusual patterns before a failure occurs. Computer-vision systems can assist with aircraft inspections, runway monitoring, baggage handling, and airport security. Airlines use AI to forecast passenger demand, plan schedules, answer customer questions, and manage ticket prices. Air traffic organizations are examining AI for traffic-flow prediction, conflict detection, and more efficient use of airspace.

The European Union Aviation Safety Agency has developed a human-centered AI roadmap that addresses aircraft operations, production, maintenance, air traffic management, airports, drones, cybersecurity, environmental concerns, and safety-risk management. EASA has also created guidance for AI systems that enhance human abilities and for systems that may make decisions under human oversight. Its guidance emphasizes learning assurance, explainability, ethics, and safe human-AI interaction (European Union Aviation Safety Agency [EASA], 2023, 2024).

The FAA has also developed an AI safety-assurance roadmap. The roadmap recommends introducing AI incrementally, using existing aviation safety practices, clearly assigning responsibility, and treating AI as a tool rather than as a human. It identifies collaboration, workforce readiness, safety research, AI assurance, and the use of AI to improve aviation safety as major priorities. The FAA also recognizes that the aviation industry still needs dependable methods for demonstrating that AI systems are safe enough for aircraft and operational use (FAA, 2024).

These existing systems provide a foundation for AGI, but they remain separated by task, company, aircraft type, or department. The proposed AGI application would connect these separate areas while preserving human authority.

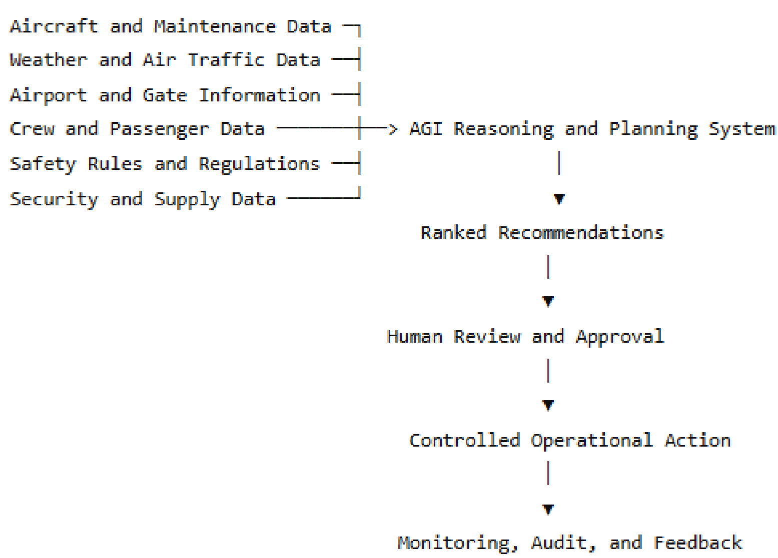
AGI Application Proposal: Aviation General Intelligence Operations Partner

My proposal is an Aviation General Intelligence Operations Partner, or AGIOP. This system would serve as a decision-support platform for airlines, airports, maintenance organizations, and air traffic professionals. It would not replace pilots, mechanics, dispatchers, controllers, or safety managers. Instead, it would collect approved information from different aviation systems, understand how the information is connected, evaluate possible actions, and explain its recommendations to authorized employees.

The AGIOP system would receive information from aircraft sensors, maintenance records, weather services, air traffic systems, airport operations, crew schedules, passenger connections, safety reports, and supply-chain databases. It could examine a developing situation and create several response plans. Each plan would include expected safety effects, delays, costs, fuel use, passenger impact, staffing needs, and regulatory limitations. Human decision-makers would review and approve the plan before action was taken.

Figure 1

Proposed Human-Centered AGI System



One major use would be disruption management. Consider an aircraft that develops a mechanical problem while severe weather approaches an airport. Current systems may separately identify the mechanical warning, predict the storm, calculate crew-duty limits, or search for another aircraft. AGIOP could combine all of these factors. It could recommend whether to repair the aircraft, replace it, delay the flight, cancel it, change gates, reroute passengers, or move the departure to another airport. It could also explain why one option produces a lower safety risk or fewer passenger disruptions.

A second use would be predictive maintenance and parts management. AGIOP could examine aircraft sensor readings, inspection images, previous repairs, manufacturer information, regulatory directives, mechanic notes, and the availability of replacement parts. Instead of only predicting that a component may fail, it could determine when and where the aircraft should be serviced, identify qualified personnel, reserve a maintenance location, order the correct part, and reorganize the flight schedule. Licensed mechanics and maintenance supervisors would still conduct inspections and approve the aircraft's return to service.

A third use would be air traffic and environmental planning. The system could compare weather, congestion, airport capacity, aircraft performance, noise restrictions, and fuel requirements. It could recommend routes that reduce delays and unnecessary fuel use while maintaining required safety margins. During unusual events, such as widespread thunderstorms or an airport closure, the system could quickly produce plans that consider the entire network instead of improving one flight while creating problems for many others.

A fourth use would involve training and workforce support. AGIOP could create personalized training scenarios for pilots, mechanics, dispatchers, flight attendants, airport employees, and controllers. It could identify areas where an employee needs more practice and generate realistic simulations based on safety reports and operational trends. The system could also help employees locate regulations, maintenance procedures, or emergency guidance. However, official manuals and approved procedures would remain the controlling sources.

Finally, AGIOP could improve passenger accessibility and communication. During a disruption, it could provide consistent information in different languages and accessible formats. It could consider connecting flights, mobility assistance, unaccompanied minors, medical accommodations, and baggage locations when recommending rebooking options. Human customer-service employees would handle unusual or sensitive situations.

Anticipated Benefits

The first expected benefit is improved efficiency. Instead of waiting for several departments to exchange information and create separate plans, AGIOP could identify connections immediately. This could reduce delays, aircraft downtime, repeated work, and communication errors.

A second benefit is stronger safety management. The system could examine large numbers of safety reports and operational records to detect patterns that may be difficult for a person to identify. It could compare new events with previous incidents, unusual sensor readings, maintenance findings, and operational conditions. The FAA has recognized that data mining and clustering may help uncover outliers, unexpected situations, and information that traditional analysis could miss (FAA, 2024).

A third benefit is improved innovation. Because an AGI system could transfer knowledge between areas, lessons from maintenance could improve training, and lessons from air traffic disruptions could improve airline scheduling. The system could test possible procedures in simulations before they are used in actual operations.

AGIOP could also improve accessibility and sustainability. Better flight planning may reduce fuel waste, while improved passenger support could make air travel easier for travelers with disabilities, language barriers, or complex connections. However, these benefits would depend on the quality of the data and the controls placed around the system.

Risks and Ethical Concerns

The greatest concern is safety. An incorrect AGI recommendation could affect an aircraft, airport, or a large part of the air transportation network. The system could misunderstand incomplete information, create a plan that violates a regulation, or produce a recommendation that appears reasonable but contains a hidden error. For this reason, AGIOP should begin as an advisory system and should not independently control aircraft or issue air traffic instructions.

Explainability would also be necessary. Pilots, mechanics, controllers, dispatchers, and regulators must understand why the system made a recommendation. A statement such as "the computer selected this option" would not be acceptable. The system should show the information it used, the rules it followed, the alternatives it considered, its level of confidence, and any uncertainties.

Privacy is another concern. Passenger records, employee information, security data, and aircraft operational data must be protected. The AGI system should receive only the information required for an approved task. Sensitive data should be encrypted, access should be based on job responsibilities, and every use of the system should be recorded.

Bias could affect decisions involving passengers or employees. Historical data may contain unequal treatment or may not represent all airports, communities, aircraft, or operating conditions. Human reviewers should examine whether recommendations unfairly disadvantage certain groups. Data should also be tested for accuracy, completeness, and representation before it is used.

Cybersecurity would be especially important because aviation systems are critical infrastructure. An attacker might attempt to change data, steal information, give the AGI false instructions, or interfere with its recommendations. AGIOP should be separated from direct aircraft-control systems, use strong identity verification, operate under least-privilege access, and have independent backup systems. Employees must also be able to safely stop using the AGI and return to approved manual procedures.

Workforce concerns must also be addressed. Some employees may fear that AGI will replace their positions. The better approach is to use AGI to support employees rather than remove them. Aviation organizations should retrain workers for roles involving AI oversight, data quality, system safety, cybersecurity, simulation, and investigation. Final legal and professional responsibility should remain with clearly identified people and organizations.

The NIST AI Risk Management Framework provides a useful structure for controlling these risks. Its four major functions—Govern, Map, Measure, and Manage—could guide the system throughout its life cycle. Aviation organizations should establish responsibility, identify possible harms, test system performance, respond to discovered risks, and continuously monitor the system after deployment (National Institute of Standards and Technology [NIST], 2023).

Feasibility and Implementation

AGIOP should be introduced in stages. The first stage would involve low-risk administrative functions, such as document searches, training support, and summarizing operational information. The second stage could include maintenance planning, passenger rebooking recommendations, and airport resource scheduling. Later stages could support flight dispatchers, safety analysts, and air traffic-flow managers.

Before each stage, the system should be tested in simulations and compared with decisions made by experienced professionals. Independent evaluators should examine accuracy, cybersecurity, bias, explainability, and performance during unusual conditions. Aviation regulators, industry organizations, employee representatives, researchers, and passengers should participate in developing standards.

This gradual approach follows the FAA's recommendation to begin with lower-risk uses, gain experience, and expand only when acceptable safety evidence is available. It also supports EASA's human-centered model in which advanced systems operate under meaningful human oversight.

Conclusion

Artificial General Intelligence could transform aviation by connecting information and decisions that are currently divided among different departments and specialized systems. A human-centered Aviation General Intelligence Operations Partner could support predictive maintenance, disruption management, air traffic planning, sustainability, training, and passenger accessibility. Its greatest value would be its ability to reason across multiple areas and respond to unfamiliar situations.

However, aviation should not adopt AGI simply because the technology becomes available. Safety, cybersecurity, privacy, fairness, explainability, workforce preparation, and legal responsibility must be addressed before advanced systems receive operational authority. AGI should first serve as a carefully controlled partner that provides recommendations while trained aviation professionals remain responsible for important decisions. With gradual implementation, independent testing, strong regulation, and meaningful human oversight, AGI may help create an aviation system that is safer, more efficient, accessible, resilient, and environmentally responsible.

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